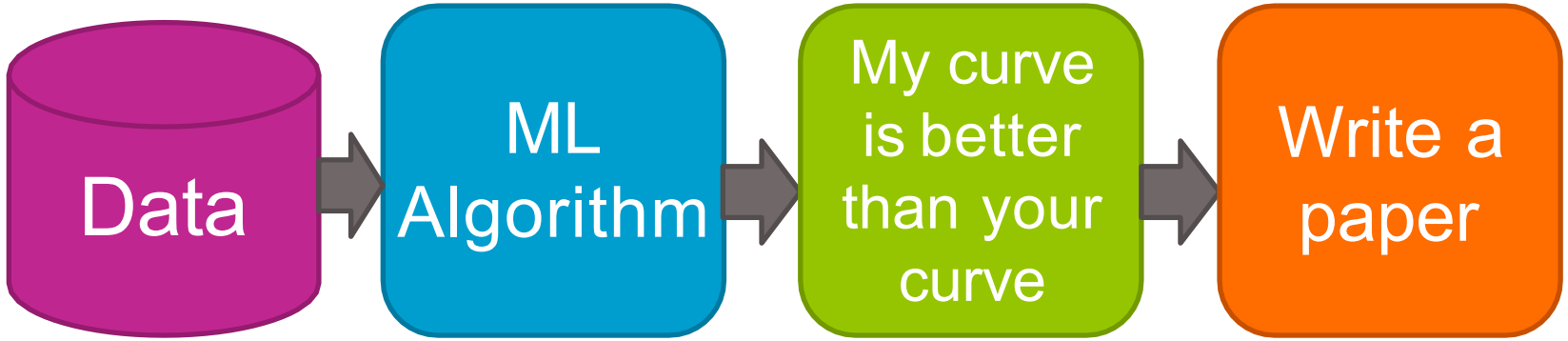
Artificial Intelligence

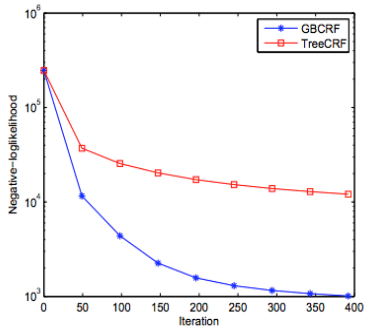
Dr. Tabrej Khan

Machine learning is changing the world

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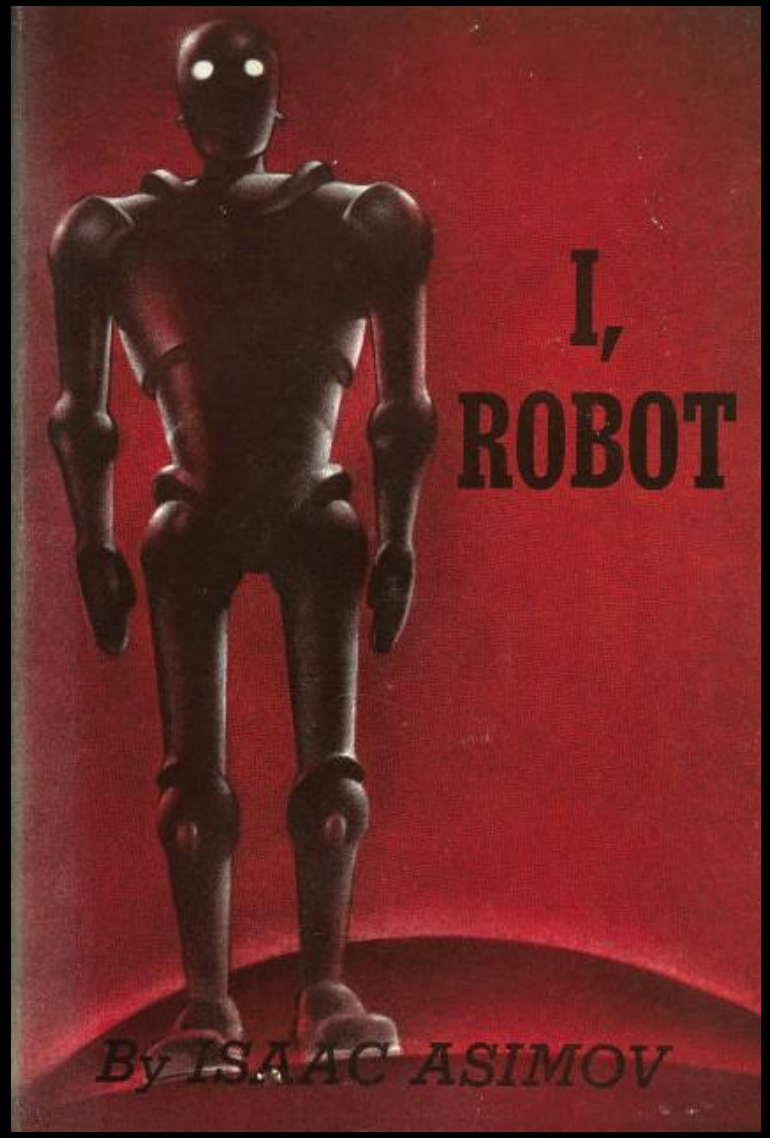
Old view of ML

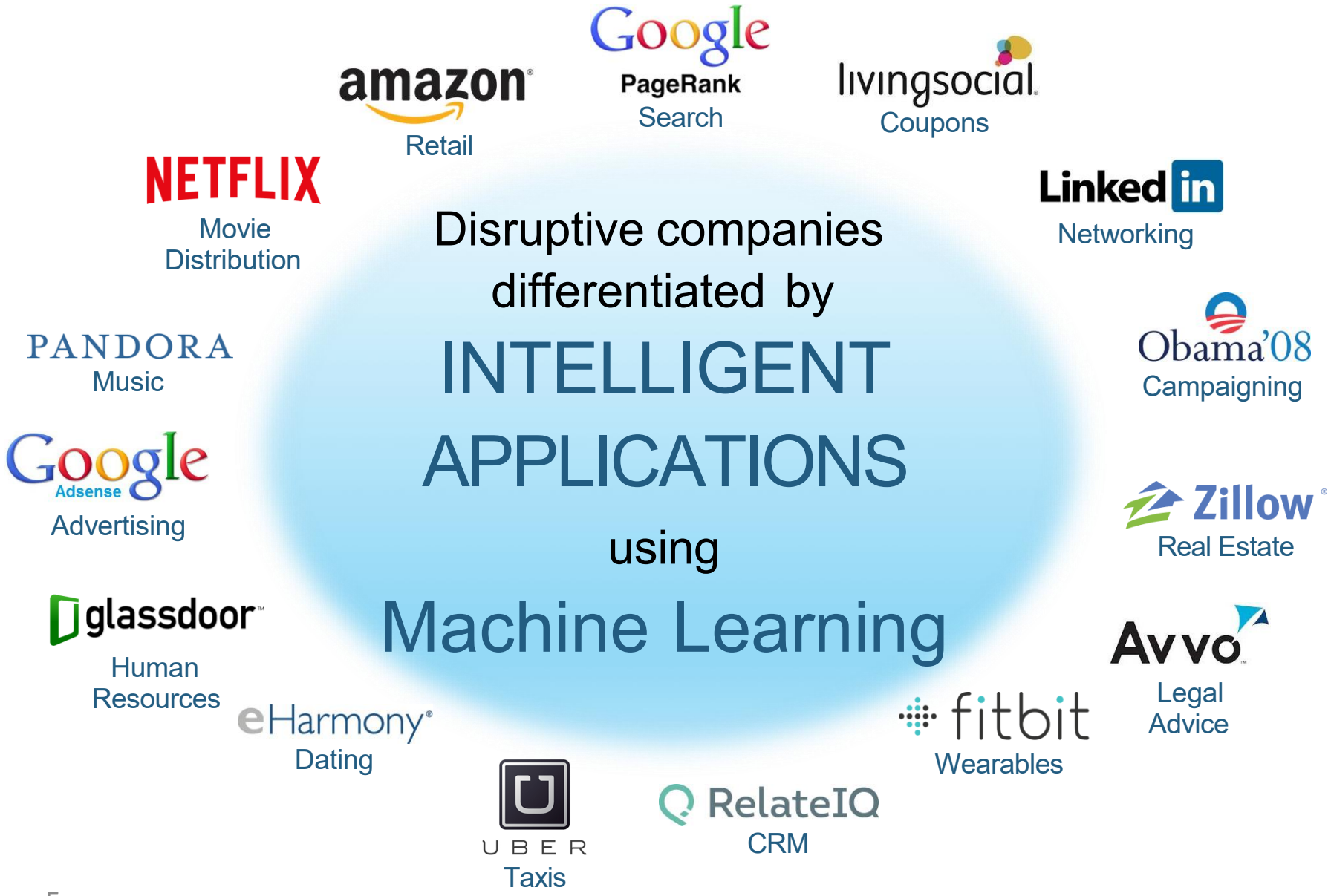




Neural Information Processing Systems Foundation

ICML


I,
ROBOT
By ISAAC ASIMOV


Google
PageRank
Search
amazon
Retail
livingsocial
Coupons
NETFLIX
Movie
Distribution
LinkedIn
Networking
Disruptive companies
differentiated by
INTELLIGENT
APPLICATIONS
using
Machine Learning
PANDORA
Music
Obama'08
Campaigning
Google
Adsense
Advertising
Zillow
Real Estate
glassdoor
Human
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Wearables
UBER
Taxis
RelateIQ
CRM

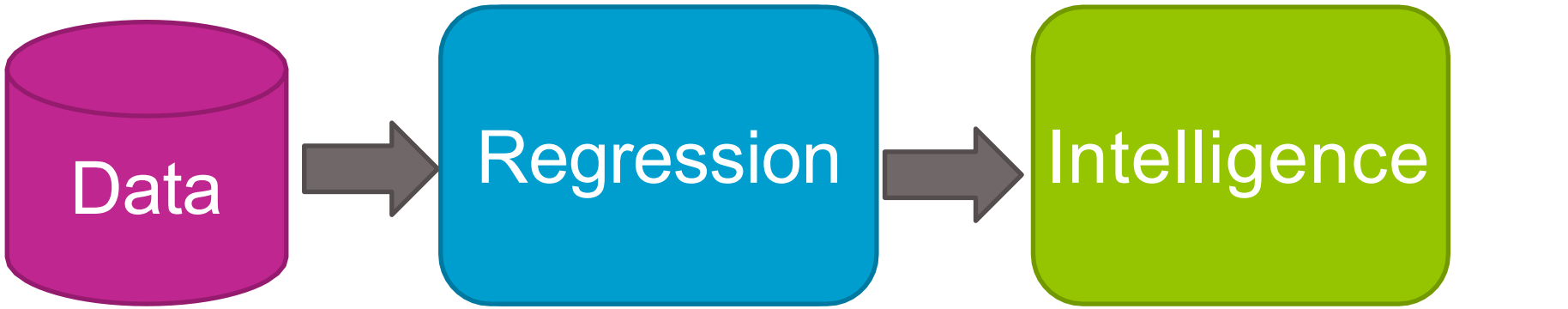
The machine learning pipeline



ML case studies

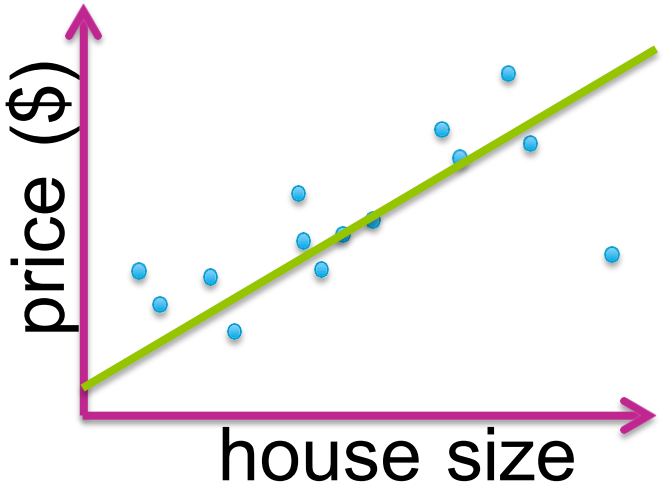
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Case Study 1: Predicting house prices





+ house features



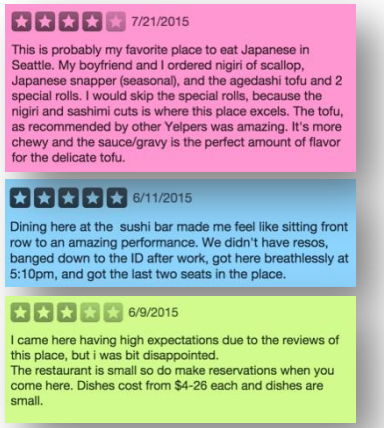


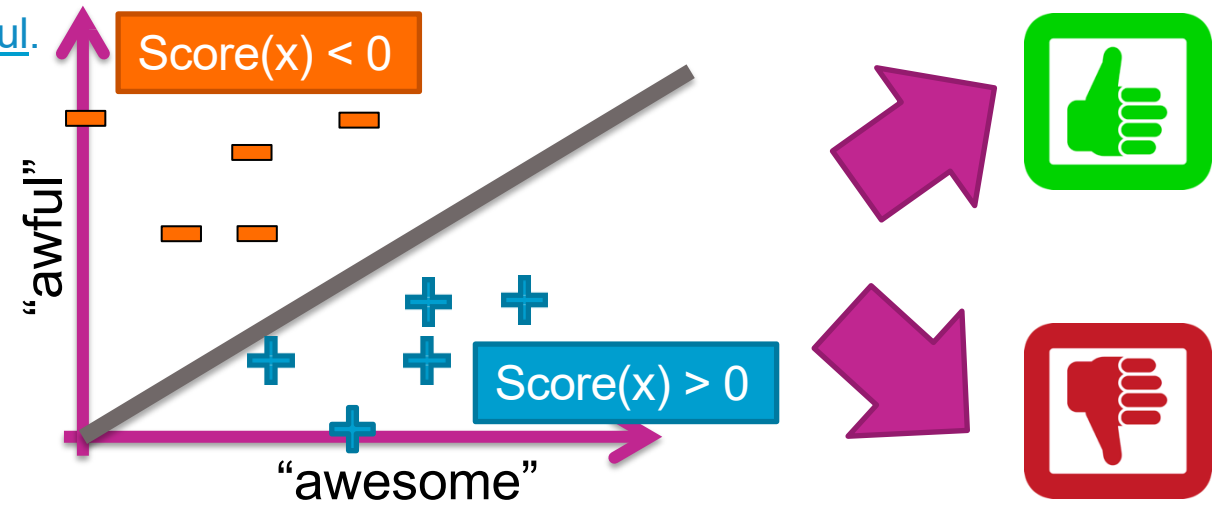
Case Study 2: Sentiment analysis



Sushi was awesome,
the food was awesome,
but the service was awful.

All reviews:





Case Study 3: Document retrieval



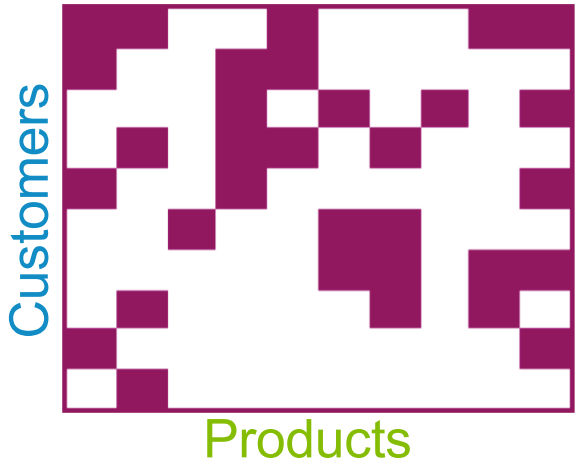
Case Study 4: Product recommendation



Your past purchases:



+ purchase histories of all customers



Recommended items:



Case Study 4: Product recommendation



Your past purchases:



+ purchase histories of all customers

Customers



features

features

features

Products



features

features

features

Recommended items:

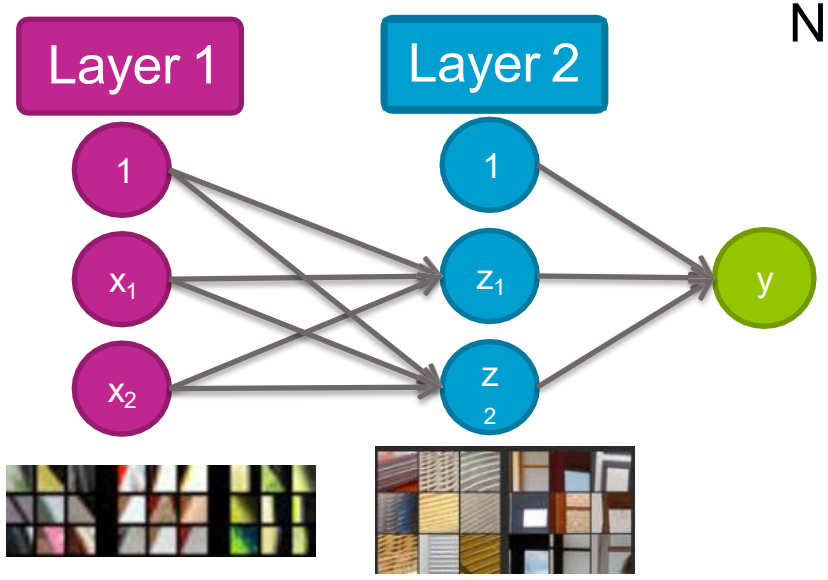


Case Study 5:
Visual product recommender



Input images:





Nearest neighbors:



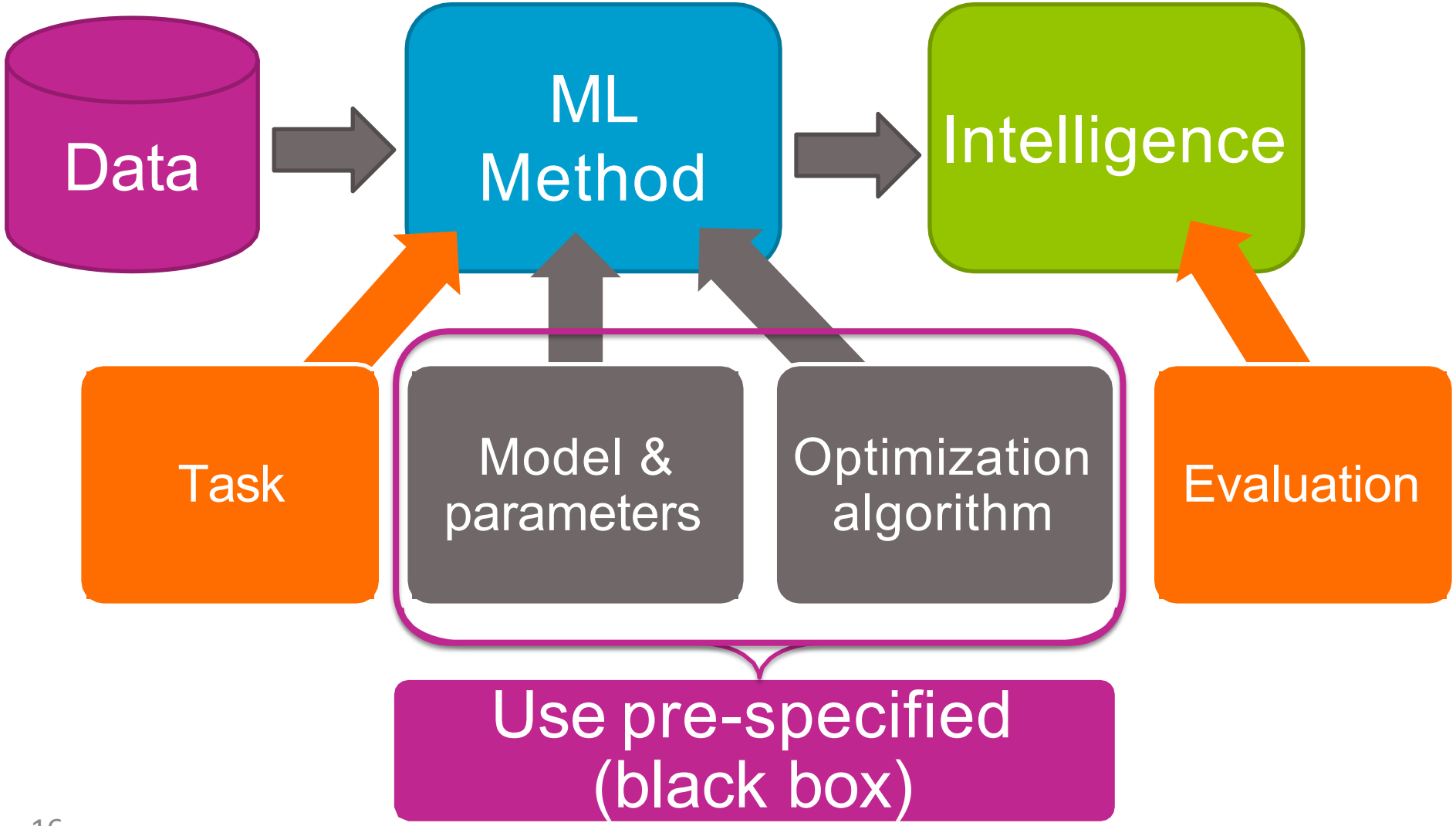
A unique ML specialization

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Not like other ML courses out there…

From use cases to models & algorithms

This course is about building, evaluating and deploying *intelligence in each case study*…



Subsequent this courses provide depth in models & algorithms, but still use case studies

2. Regression
3. Classification
4. Clustering & Retrieval
5. Matrix Factorization & Dimensionality Reduction
6. *Capstone:* Build an Intelligent Application with Deep Learning

2. Regression

Case study: Predicting house prices

Models

- Linear regression
- Regularization:
  Ridge (L2), Lasso (L1)

Algorithms

- Gradient descent
- Coordinate descent

Concepts

- Loss functions, bias-variance tradeoff, cross-validation, sparsity, overfitting, model selection

3. Classification

Case study: Analyzing sentiment

Models

- Linear classifiers (logistic regression, SVMs, perceptron)
- Kernels
- Decision trees

Algorithms

- Stochastic gradient descent
- Boosting

Concepts

- Decision boundaries, MLE, ensemble methods, random forests, CART, online learning

4. Clustering & Retrieval

Case study: Finding documents

Models

- Nearest neighbors
- Clustering, mixtures of Gaussians
- Latent Dirichlet allocation (LDA)

Algorithms

- KD-trees, locality-sensitive hashing (LSH)
- K-means
- Expectation-maximization (EM)

Concepts

- Distance metrics, approximation algorithms, hashing, sampling algorithms, scaling up with map-reduce

5. Matrix Factorization & Dimensionality Reduction

Case study: Recommending Products

Models

- Collaborative filtering
- Matrix factorization
- PCA

Algorithms

- Coordinate descent
- Eigen decomposition
- SVD

Concepts

- Matrix completion, eigenvalues, random projections, cold-start problem, diversity, scaling up

6. *Capstone:*

An intelligent application using deep learning

Build & deploy
a recommender using
product images and
text sentiment

This specialization is for you if…

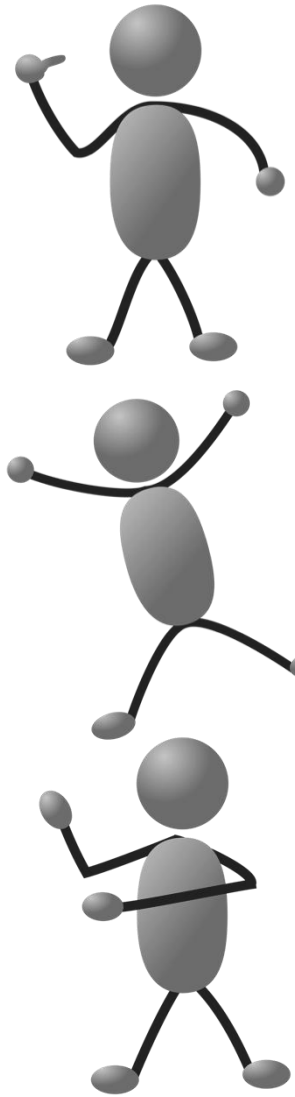
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Level of the specialization

Motto:
tough concepts made intuitive
and applicable

minimize prereq knowledge
maximize ability to develop and deploy
learn concepts through case studies

Target audience

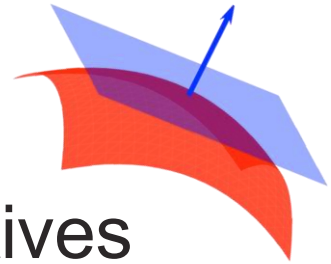


Software engineer

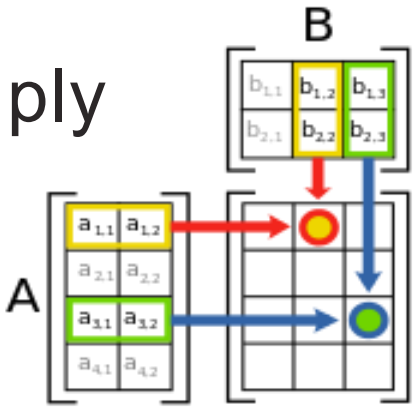
Scientist

Data enthusiast

Math background

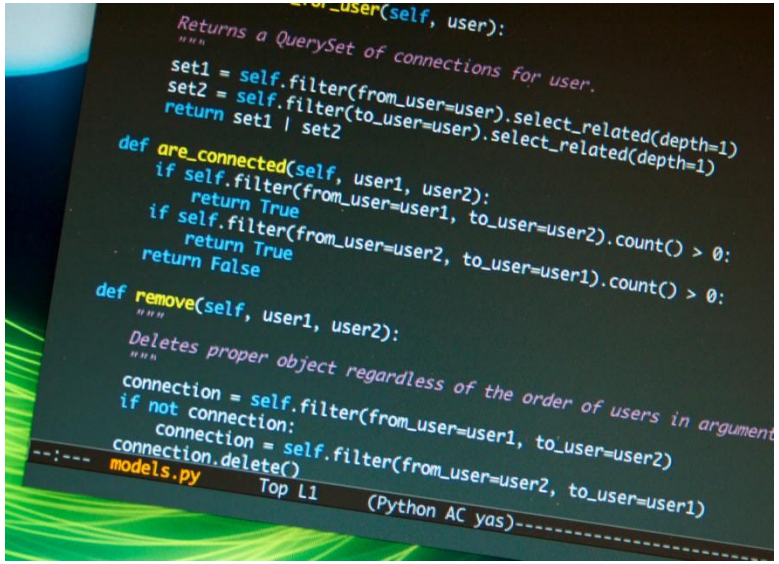


- Basic calculus
  - Concept of derivatives
- Basic linear algebra
  - Vectors
  - Matrices
  - Matrix multiply



Programming experience

- Basic Python used
  - Can pick up along the way if knowledge of other language





Computing needs

- Basic desktop or laptop
- Access to internet
- Ability to:
  - Install and run Python
  - Store a few GB of data



You'll be able to do amazing things…

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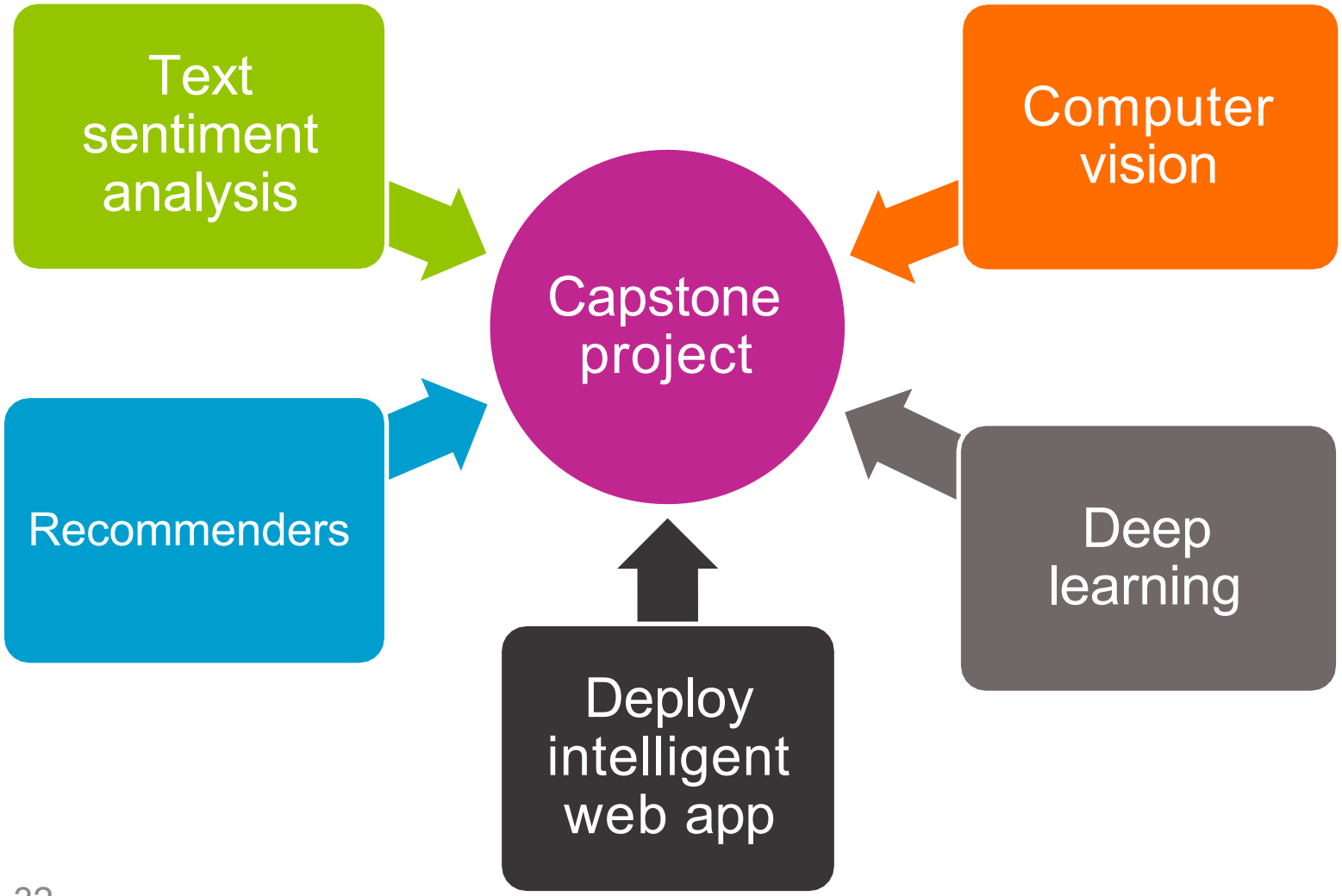
Project:
Build and deploy an intelligent application with deep learning

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An intelligent recommender using images & text

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We will do something even more exciting…



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